

Submission DSA1-FINAL-SUB07

Question DSA1-FINAL-Q01

beginning only... show step

$7 > 4$, go right; $7 < 9$, go left; $7 > 6$, go right.

[blank space]

I did not reach a conclusion

Question DSA1-FINAL-Q02

beginning only... show step

Maintain a hash set of values seen so far.

[blank space]

I did not reach a conclusion

Question DSA1-FINAL-Q03

beginning only... show step

A balanced tree can be built recursively by choosing the median, with each key visited once: $\Theta(n)$.

[blank space]

I did not reach a conclusion

Question DSA1-FINAL-Q04

beginning only... show step

Queue A; discover B,C; then dequeue B and discover D; dequeue C and discover E.

[blank space]

I did not reach a conclusion

Question DSA1-FINAL-Q05

beginning only... show step

Invariant: the prefix before index i is sorted and contains the original prefix values.

[blank space]

I did not reach a conclusion

Question DSA1-FINAL-Q06

beginning only... show step

$7 > 4$, go right; $7 < 9$, go left; $7 > 6$, go right.

[blank space]

I did not reach a conclusion

Question DSA1-FINAL-Q07

beginning only... show step

Maintain a hash set of values seen so far.

[blank space]

I did not reach a conclusion

Question DSA1-FINAL-Q08

beginning only... show step

A balanced tree can be built recursively by choosing the median, with each key visited once: $\Theta(n)$.

[blank space]

I did not reach a conclusion

Question DSA1-FINAL-Q09

beginning only... show step

Queue A; discover B,C; then dequeue B and discover D; dequeue C and discover E.

[blank space]

I did not reach a conclusion

Question DSA1-FINAL-Q10

beginning only... show step

Invariant: the prefix before index i is sorted and contains the original prefix values.

[blank space]

I did not reach a conclusion